

SAI SUSHANT KORANNE

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EDUCATION

DePaul University

Master of Science in Game Programming (STEM), CDM

Chicago, IL

2025-2027

Relevant Courses: Optimized C++, Programming Design patterns, SOLID principles, Game Engine Architecture, 3D-Math and Liner Algebra, Game Physics, Multithreading, CUDA

Vellore Institute of Technology

Bachelor of Technology in Computer Science Engineering

Vellore, India

2017-2021

SKILLS

- **Game Engines:** Unity, Unreal Engine
- **Programming Languages:** C#, C++, Java, C
- **VCS:** Git, Helix-P4V
- **OS:** Windows, Linux, Mac
- **IDE/Code Editors:** Visual Studio, VS Code, JetBrains Rider

EXPERIENCE

SuperHuge Studio-Xarpie Solutions

Game Developer

Bengaluru, India

November 2022 – June 2024

- **Gameplay & Mechanics:** Partnered with cross-functional teams to iterate on prototypes, define gameplay mechanics, and deliver cohesive player experiences from concept to final release.
- **System Architecture & Design:** Architected scalable core game systems and logic in Unity (C#), establishing robust design patterns to ensure modularity and highly maintainable codebases across 2D and 3D projects.
- **Quality & Process Management:** Streamlined QA workflows by leading bug triage and regression testing; standardized version control and technical documentation to accelerate team onboarding.

Cognizant Technology Solutions Pvt. Ltd

Programmer Analyst

Hyderabad, India

August 2021 - November 2022

- **DevOps & Pipeline Automation:** Architected CI/CD pipelines and managed Git branching strategies to automate builds, testing, and stable cross-environment deployments.
- **Infrastructure & Containerization:** Built and maintained Docker images for consistent runtime environments; automated local and server setups utilizing PowerShell scripts to drastically reduce onboarding time.
- **Release Lifecycle:** Administered application servers to maximize uptime, managed agile product backlogs, and coordinated with stakeholders to ensure the timely delivery of prioritized features.

PROJECTS

Game Engine (Using Dx11)

- Engineered a custom C++/DirectX 11 engine featuring a GLTF/Protobuf asset pipeline, proprietary math library (Transform matrix and Quaternions), and a PCS tree for hierarchical scene graph management.
- Implemented advanced rendering pipelines utilizing compute shaders for high-performance processing and GPU-accelerated vertex skinning for playback of skeletal animations.

Space Invaders (Custom Game Engine - C#)

- Architected a custom C# 2D engine utilizing Factory, Observer, and State design patterns to create a scalable, modular gameplay framework.
- Developed real-time collision detection and optimized update loops to ensure highly efficient entity management and smooth performance.

Audio Engine (Multithreading)

- Built a low-latency C++/XAudio2 engine utilizing a robust five-thread architecture to handle parallel I/O, playback control, and low-level processing.
- Implemented dynamic priority scheduling and user-defined event hooks to enable seamless, asynchronous real-time audio streaming.

Obstacles Path (Unreal)

- Created a dynamic 3D traversal game in Unreal requiring precision timing to navigate rotating, sweeping, and vertically moving hazards.
- Designed a goal-oriented progression system centered around real-time player reaction to complex, synchronized obstacle patterns.

Multiplayer Top Spin Game (Unity)

- Engineered a real-time multiplayer arena in Unity using Photon PUN2 and RPCs for authoritative matchmaking and precise player synchronization.
- Designed physics-driven combat where momentum-based collisions dynamically reduce rotational speed to trigger player elimination.

Racing Game Using Reinforcement Learning (Unity)

- Developed a high-speed racing game featuring AI agents trained via reinforcement learning to dynamically adapt flight tactics against the player.
- Formulated a checkpoint-based reward and penalty system to train efficient, competitive AI navigational policies.

Maniacs (Unity)

- Programmed a survival-horror game featuring dynamic enemy AI capable of navigating procedurally darkened environments to pursue the player.
- Implemented distinct melee/ranged weapon systems and balanced resource management mechanics to drive tense, tactical gameplay.